

SonoSPADE: Real-Time, Per-Tissue Ultrasound Texture Synthesis for Closed-Loop Acquisition

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Abstract. A closed-loop simulator for learning ultrasound *acquisition* renders a new B-mode frame at every agent step, so its texture stage must be fast and controllable. Realism-first diffusion is far too slow to run in the loop, and global sim-to-real translators use a single appearance map across the whole image, giving no per-tissue control. We present *SonoSPADE*, a per-tissue ultrasound texture stage that is real-time and GPU-free, trained unpaired with no real labels. The generator is conditioned on *both* a physics render and the tissue label slice. A segmenter, pretrained for free on aligned simulator pairs and then frozen, pseudo-labels the real pool for an OASIS-style discriminator and drives a per-tissue consistency loss. A single pass runs at 15 to 85 frames per second *without a GPU*, two to three orders of magnitude faster than diffusion. On the public Kaggle abdominal set, when we change a region’s tissue label, only that region’s texture changes, while methods that ignore the label do not react at all (label-swap locality ≈ 26 , versus 0). Per-organ intensity matches real about $2\times$ better than any learned baseline, and structure is preserved $2.4\times$ better; on whole-frame FID/KID it trails by design. As a first in-the-loop proof of concept, on an organ-conditioned task scored by a *fixed* real-trained recognizer shared across texture stages, the per-tissue reward gives the highest ground-truth liver visibility in all three seeds. Code and the reproducible pipeline are released.

Keywords: Ultrasound synthesis · Semantic image synthesis · Real-time synthesis · Closed-loop simulation · Sim-to-real.

1 Introduction

We study how to learn ultrasound *acquisition* in simulation: an agent moves a virtual probe and observes the B-mode frame it would produce, so a texture stage must render a new frame at every step. The stage must be fast, must run on an ordinary computer, and must be controllable, so that the agent’s actions map to anatomy. Realism-first synthesizers do not meet these needs. Diffusion produces high-quality ultrasound but takes seconds per frame [7], far too slow for a loop that uses millions of frames. Global sim-to-real translators work on the grayscale image alone [6], applying one appearance map to the whole frame, so they cannot make sure that liver reads as liver and kidney as kidney. Yet

the pipeline already knows the anatomy: a canonical label slice drives a physics B-mode with acoustic shadowing, a depth-varying point spread function, and correlated speckle. We make the learned stage *per tissue* by conditioning it on that label slice, and keep it single-pass so it runs in real time without a GPU. This improves on both semantic image synthesis [4] and segmentation-consistent CT-to-US translation [6]; realism-first anatomy-consistent synthesizers [11] aim at offline augmentation at higher fidelity, while we trade fidelity for the speed and per-tissue control a texture stage needs in the loop.

Two design choices set SonoSPADE apart. First, the generator is conditioned on *both* the physics render and the label, so it refines a tissue-correct render instead of inventing ultrasound from CT. Second, a frozen segmenter, pretrained for free on aligned pairs, supplies both the OASIS-style discriminator’s pseudo-labels [8] and a per-tissue consistency term, removing SPADE’s need for paired real data [4]. A one-sided contrastive backbone [3] keeps training on a laptop without CUDA. Our contributions are:

- A real-time, GPU-free per-tissue texture stage: a single pass at 15–85 frames/s on a laptop with no discrete GPU, two to three orders of magnitude faster than diffusion, that runs in the loop without becoming the bottleneck.
- A physics-conditioned dual-input generator, conditioned on *both* the physics render and the label slice, giving per-tissue control (relabel a region and only its texture changes) with free unpaired segmentation consistency, so no real labels are needed.
- A per-tissue evaluation (per-class Wasserstein, a label-swap locality metric, speckle SNR/GLCM), plus a diagnosis and two fixes (a sector remap and test-time adaptation) that move SonoSPADE from worst to ahead of the raw render on downstream transfer.
- A first in-the-loop proof of concept: under a *fixed* real-trained recognizer shared across texture stages, the per-tissue reward gives the highest ground-truth liver visibility in all three seeds.

2 Related Work

Semantic synthesis and label-consistent US. SPADE conditions generation on a label map via spatially-adaptive normalization [4], and OASIS recasts the critic as a per-pixel segmentation discriminator [8]; both are supervised and label-only. We borrow the adaptive conditioning but drive it from a physics render and supervise the discriminator with *free* pseudo-labels. A parallel line generates label-consistent US from anatomical models to train segmenters (e.g., a CycleGAN at 87–91 Dice on cardiac structures [1]); these whole-image translators offer no per-tissue control, which we expose by conditioning on the label.

Anatomy-consistent US synthesis. Closest to us, Vitale et al. [11] combine a segmentation-guided loss and polar (sector) training to suppress hallucinations and improve realism; related structure-preserving translators include S-CycleGAN [6,14] and CACTUSS [9]. These optimize offline realism; we optimize per-frame latency for in-the-loop use, and add per-tissue *conditioning* (not

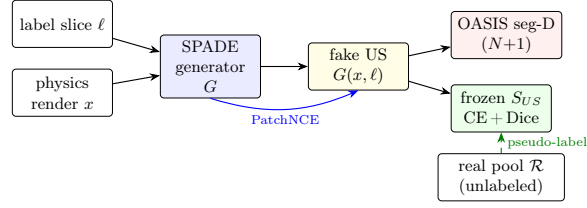


Fig. 1. SonoSPADE. The label slice and the physics render both condition the generator. A frozen segmenter S_{US} , pretrained free on aligned simulator pairs, pseudo-labels the real pool for an OASIS ($N+1$) discriminator and enforces a per-tissue consistency loss; PatchNCE preserves content. No inverse generator, no cycle loss.

merely a loss), a label-swap locality metric, and single-pass GPU-free inference. Semantic diffusion [7] reaches higher fidelity but needs tens to hundreds of passes per frame, too slow for an agent that uses millions of frames.

3 Method

Fig. 1 summarizes SonoSPADE. Let x be a physics B-mode render and ℓ its one-hot canonical tissue label slice ($K=13$ classes). The generator $G(x, \ell)$ refines x into a realistic frame; it is trained unpaired against a real pool $\mathcal{R}$ with three forces.

3.1 Physics-conditioned dual-input generator

G is a ResNet encoder-decoder whose residual and upsampling blocks are SPADE-normalized: a parameter-free instance norm whose per-pixel scale and bias are predicted from the resized label map ℓ , so the semantic layout modulates activations at every scale [4]. The physics render x is the content stream, so G refines a tissue-correct render instead of hallucinating anatomy. Content is preserved by multilayer PatchNCE on G ’s own encoder features [3] between x and $G(x, \ell)$; there is no inverse generator and no cycle loss.

3.2 Free unpaired segmentation consistency

The simulator emits perfectly aligned (x, ℓ) pairs, so a compact U-Net [5] segmenter S_{US} is pretrained at no annotation cost, then *frozen* and used two ways. (i) It pseudo-labels the unlabeled real pool, giving an OASIS-style ($N+1$)-class per-pixel discriminator D a semantic target for free [8]: G wins only when each synthesized pixel reads as its conditioned tissue. (ii) A consistency loss $\mathcal{L}_{seg} = \text{CE} + \text{Dice}$ requires $S_{US}(G(x, \ell))$ to match ℓ , so G cannot collapse to a global recolor. The full objective is

$$\mathcal{L} = \mathcal{L}_{adv}^{\text{OASIS}} + \lambda_{nce}\mathcal{L}_{nce} + \lambda_{seg}\mathcal{L}_{seg} + \lambda_{tc}\mathcal{L}_{tc}, \quad (1)$$

where $\mathcal{L}_{adv}$ uses class-balanced per-pixel cross-entropy, so rare tissues are not ignored.

3.3 Grouped scheme and training

Fat, GI wall, pancreas, muscle, and generic soft tissue have near-equal backscatter and render almost identically, so S_{US} and D use a *grouped* scheme (these five merge; distinct structures keep their class) while G still conditions on the full K classes. Three light stabilizers help: a temporal-consistency term $\mathcal{L}_{tc}$ requiring G to commute with a small in-plane shift, an OASIS-style LabelMix on D [13], and an R1 gradient penalty [2]. The deployed generator is an EMA of the training weights.

4 Experiments

4.1 Setup

We build on an ultrasound substrate that runs on an ordinary laptop: it renders a physics B-mode from a CT-derived canonical label volume (13 tissues via TotalSegmentator [12]) and exposes a label-conditioned translator interface. Free aligned (x, ℓ) pairs come from eight abdominal CT cases with pose sweeps and domain randomization. For the real head-to-head, we use the public Kaggle abdominal ultrasound set [10]: every translator is trained on its 404-frame real train split and evaluated on the disjoint 60 annotated real frames (liver, kidney, gallbladder, spleen, vessels), a leakage-free real test on the same held-out content. Everything runs on Apple Silicon (MPS, no CUDA).

4.2 The label conditioning is genuinely per-tissue

The central claim is that texture is per-tissue, not a global recolor. Holding the physics render fixed, we relabel a tissue region and measure the texture change inside versus outside. A global recolor scores near zero inside. SonoSPADE changes the region’s own texture by 0.065 inside versus 0.005 outside, a *pooled* inside-to-outside ratio of about 13. The label-swap locality metric in Table 1 is a stricter per-class summary: it averages each tissue’s own inside/outside ratio, giving **26.3** for SonoSPADE versus 0 for every label-free translator by construction (supplementary Fig. S1). The per-class mean exceeds the pooled 13 because tissues with very little out-of-region change give large ratios. Per-class speckle SNR also shifts per tissue, confirming statistics learned per tissue, not one uniform map.

4.3 Free segmentation supervision

Pretrained on the aligned pairs, S_{US} reaches 0.93/0.90/0.72 Dice on liver/lung/vessels (acoustically distinct) on held-out patients. The canonical 13-class macro Dice is capped at 0.50 by indistinct soft tissues; merging them in the grouped scheme lifts it to 0.66 (merged soft-tissue class 0.75 versus 0.35), with no change to distinct tissues. This reflects B-mode acoustics, not a tuning trick.

Table 1. Baseline comparison on the real Kaggle set. Edge IoU (structure vs. physics input, higher better); *organ* intensity W_1 (mean over the five annotated organs, excluding background, lower better); label-swap locality (0 for label-free translators by construction, higher better); TSTR Dice. SonoSPADE preserves structure $2.4\times$ better and per-organ intensity about $2\times$ better than any learned translator, and is the only method with per-tissue control. The SonoSPADE-curvi row is the same content in the matched sector geometry (Sec. 4.5); its TSTR is the five-seed mean from Table 2.

Method	Edge IoU $\uparrow$	Organ W_1 $\downarrow$	Locality $\uparrow$	TSTR Dice $\uparrow$
Raw physics	1.00	0.064	0.0	0.049
CycleGAN	0.11	0.266	0.0	0.032
CUT	0.17	0.246	0.0	0.031
S-CycleGAN (reimpl.)	0.13	0.223	0.0	0.026
SonoSPADE	0.41	0.119	26.3	0.025
SonoSPADE-curvi	0.54	0.115	18.6	0.099

4.4 Comparison to baselines

Table 1 compares raw physics, CycleGAN [14], CUT [3], an S-CycleGAN-style variant [6], and SonoSPADE on the real Kaggle set. **Structure:** SonoSPADE preserves anatomy far better than any learned translator (edge IoU 0.41 versus 0.17/0.11 for CUT/CycleGAN, a $2.4\times$ margin): the label-free translators remap the frame to match appearance and drift on structure, while physics-plus-label conditioning keeps anatomy intact. **Per-organ realism:** We report intensity W_1 *restricted to the annotated organ masks*, since the whole-frame score is dominated by the background. Inside the masks, the recolors lose per-tissue fidelity (organ W_1 0.266/0.246 versus 0.064 for the render they started from), while SonoSPADE keeps it (0.119, about $2\times$ better than any learned baseline) and is the only method with per-tissue control (locality 26.3 versus 0). TSTR is near zero for all linear methods (SonoSPADE lowest, 0.025), dominated by a linear-versus-curvilinear geometry gap and further depressed for SonoSPADE because its conditioning shifts features from what the classifier expects; matching the geometry and adding test-time adaptation reverses the ranking (next).

Whole-frame realism (FID/KID). SonoSPADE does *not* win whole-frame FID/KID (supplementary Table S1): its scores sit near the untranslated render, above the label-free translators. FID/KID score *overall* appearance, which a global tone map matches cheaply and which is blind to whether each organ reads correctly. We do not optimize FID.

4.5 Ablation study

Ablating SonoSPADE’s terms one at a time (retrained at a fixed budget; supplementary Table S2), the segmentation-consistency term is the largest single contributor: removing it roughly halves locality (27.1 to 13.7) and degrades structure

Table 2. Downstream transfer (TSTR Dice, higher better). Curvilinear and TENT rows are mean $\pm$ std over 5 seeds; the linear row is the single run from Table 1. Geometry (A) ties SonoSPADE with the raw render; TENT (B) makes it overtake on every seed.

Setting	Raw physics $\uparrow$ SonoSPADE $\uparrow$	
Linear render (Table 1)	0.049	0.025
+ curvilinear remap (A)	0.097 ± 0.006	0.099 ± 0.011
+ curvilinear + TENT (B)	0.106 ± 0.006	0.124 ± 0.007

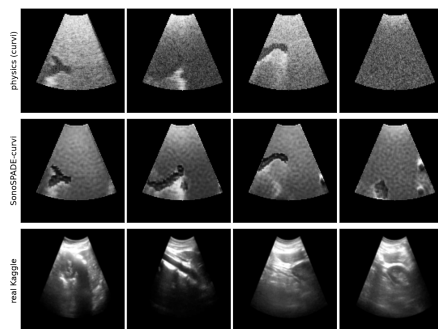


Fig. 2. Curvilinear remap (Experiment A, Sec. 4.5). After warping into the measured sector geometry, the curvilinear physics render (top), SonoSPADE-curvi (middle), and a real Kaggle frame (bottom) share the same sector field. This is the geometry match that lifts downstream transfer in Table 2.

(edge IoU 0.41 to 0.30) and per-organ W_1 (0.115 to 0.129). Locality does not fall to a label-free translator’s zero (SPADE injects the label at every scale), but the explicit consistency term roughly doubles it. Dropping the OASIS LabelMix or the temporal term leaves per-frame fidelity unchanged.

Downstream transfer: The near-zero TSTR in Table 1 is a geometry problem, not texture: our substrate renders a *linear* rectangular frame while Kaggle frames are *curvilinear* sector scans, so a segmenter trained on the linear render meets a different spatial prior at test time. Two drop-in fixes (Table 2). **(A)** A curvilinear remap warps the render (image and label) into the measured sector geometry (about 44° ; Fig. 2) at train and test time, with probe-motion augmentation: TSTR rises about $4\times$ for SonoSPADE and about $2\times$ for the raw render, tying them (0.099 ± 0.011 versus 0.097 ± 0.006 , 5 seeds). **(B)** Test-time adaptation (TENT) of the segmenter’s BatchNorm on 50 unlabeled real frames raises SonoSPADE to 0.124 but the raw render only to 0.106, so SonoSPADE *overtakes* on every one of the 5 seeds. Absolute Dice stays low, so this is directional evidence that texture aids transfer once geometry is matched, not clinical-grade segmentation.

4.6 Inference latency: real time, no GPU

Because the stage runs inside a reinforcement-learning acquisition loop, per-frame latency, not peak realism, is the limiting factor. A single generator pass takes 66 ms on CPU (15 frames/s) and 12 ms on an integrated GPU (85 frames/s) at batch one, with *no discrete GPU* (`scripts/bench_latency.py`). A diffusion synthesizer instead runs T denoising passes per frame: a representative ADM/SDM-style denoising U-Net (the family Echo-from-noise [7] builds on; 62M parameters, sized to lower-bound the cost) costs 186 ms per pass on CPU, only about $2.7\times$ a SonoSPADE pass, but its T passes make it two orders of magnitude slower *per frame* at $T=50$ ($133\times$ on CPU) and up to three at full DDPM. For millions of frames, seconds per frame is too expensive. We give up whole-frame FID/KID for this real-time, GPU-free inference, and unlike the equally fast label-free baselines, SonoSPADE is per-tissue. We verified this in place inside a MuJoCo closed-loop acquisition environment, where the texture stage is not the bottleneck. Whether it also improves *acquisition quality* is the subject of Sec. 4.7.

4.7 Per-tissue texture improves an in-the-loop acquisition agent

The latency result shows the texture stage *can* run in the loop; it does not show per-tissue texture *helps* the agent. That needs a task whose reward depends on per-tissue correctness, scored by a judge *fixed across* texture stages, not matched to each (the flaw that made our earlier goal-reaching benchmark uninformative, Sec. 5). We build one: *organ-conditioned acquisition*. A probe starts with the target organ partly in view and moves to bring it in; the per-step reward is the confidence of a frozen organ recognizer J that the organ is present. J is trained *only on real* Kaggle frames and masks (never on synthetic output, so the reward is non-circular) and is the *same* J for every texture stage. The main metric is texture-independent: the ground-truth fraction of the acquired view that is the target organ.

Is the reward usable? On held-out simulator content, J 's liver IoU is 0.48 for SonoSPADE versus 0.21 (CUT) and 0.14 (CycleGAN), approaching the real-data ceiling 0.51 (Table 3, 3 seeds); the untranslated render scores 0.27, *above* both recolors, so part of the effect is structure preservation, not per-tissue texture alone. A fair comparison also trains all three translators on the *same* real pool; an off-pool CUT checkpoint wrongly reverses the result. Trained under this reward (imitation-warm-started REINFORCE, 3 seeds, held-out patients), SonoSPADE reaches the highest ground-truth liver visibility, 0.152 ± 0.004 , above CUT (0.144), CycleGAN (0.142), and the render (0.142) from a static start of 0.11, with all three per-seed differences positive and sign-consistent (directional, not significance-tested). Its reward signal is about twice as strong (0.32 versus 0.18–0.20); the ground-truth margin is *small* because the warm-start and liver's dominance already nearly solve the task. To our knowledge, this is the first evidence that per-tissue texture helps an in-the-loop agent under a fixed, real-trained judge.

Table 3. Organ-conditioned liver acquisition (3 seeds, held-out patients). Top: a fixed real-trained recognizer J recognizes SonoSPADE’s liver far better than a recolor’s (real-data ceiling 0.51). Middle: trained under J as reward, SonoSPADE reaches the highest *ground-truth* liver visibility (static start 0.11), higher in 3/3 seeds. Bottom: its reward signal is $\sim 2\times$ stronger.

	physics	CUT	CycleGAN	SonoSPADE
J liver recognition, IoU $\uparrow$	0.27	0.21	0.14	0.48
Ground-truth liver acquired $\uparrow$	0.142	0.144	0.142	0.152
Reward signal (J conf., secondary) $\uparrow$	0.20	0.18	0.16	0.32

The effect is confined to liver: for kidney, spleen, and gallbladder *no* translator (SonoSPADE included) yields a rendering the recognizer identifies (IoU ≈ 0), though the same J scores them 0.3–0.8 on *real* data. The bottleneck is the simulator and translator failing to reproduce their appearance, not the recognizer, so the multi-organ claim rests on the offline metrics and the in-the-loop gain is shown for the organ that transfers.

5 Limitations

Unpaired training matches per-tissue statistics, not exact speckle, so the output is training texture, not diagnostic imagery, and per-tissue fidelity is strong only for tissues acoustically distinct in B-mode. We validate on a *single* dataset and anatomy (abdominal Kaggle, 60 annotated real frames, 8 CT cases); other anatomies and scanners are untested, and downstream Dice stays low even after the fixes, so transfer is directional, not clinical-grade. Our in-the-loop evidence is also bounded: it does not come from the earlier goal-reaching benchmark, whose per-texture *matched* judge is not comparable across texture stages (a random policy scores ≈ 0.36 under one judge, ≈ 0.03 under another), and the organ-conditioned gain that replaces it is small, seed-consistent rather than significance-tested (3 seeds), and liver-only.

6 Conclusion

SonoSPADE is a real-time, GPU-free, per-tissue ultrasound texture stage for closed-loop acquisition: conditioning on both a physics render and the tissue label slice makes texture per-tissue and controllable, supervised for free by a frozen simulator-pretrained segmenter with no real labels. After matching the sector geometry it overtakes the untranslated render on downstream transfer, and its per-tissue reward gives an in-the-loop agent the highest ground-truth liver visibility under a fixed real-trained judge. More broadly, a fast, controllable texture stage makes closed-loop acquisition simulation practical, a step toward training autonomous and robotic ultrasound acquisition in simulation.

The recipe trains on an ordinary laptop, and we hope it opens a research window to jointly improve realism, controllability, and speed.

Code availability. The implementation, the training and evaluation scripts, and the data preparation pipeline are available at <https://github.com/tohnperfect/SonoSpade>.

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